

should be greater than one-sixth the rise/fall time of source pulses. This phenomenon is called signal integrity (SI). Software tools can be used to assess and fix proper values of terminating resistors and preserve signal integrity at the load end.

Electromagnetic interference

In the simple circuit of a source/driver delivering a signal to the receiver in Fig. 8, let us assume that the rise/fall time is such that wires need not be treated as a transmission line. Even in this case, if the frequency of current in the loop is sufficiently high, electromagnetic radiation from the current loop will give rise to disturbance in the operation of nearby electronic circuits. The current loop will also get affected by electromagnetic radiation from nearby circuits. This phenomenon is called electromagnetic interference. The EMI generated, or susceptibility to the EMI, is proportional to the frequency of current in the loop and loop area. Hence loop area needs to be minimised to minimise EMI.

As the source and receivers are components soldered on PCBs, and connecting wires are tracks on PCBs, care is to be taken during layout stage that send and return traces are nearby in single-/two-layer PCBs. If send and return traces are far away, it will result in larger current loop area, causing EMI problems at sufficiently high frequencies. In many cases, the return trace is nothing but signal ground. In a single-/double-layer PCB, every signal line should have ground trace just by its side—it is better to have the signal trace surrounded by ground traces on either sides.

If a PCB is very dense and doesn't have space for so many signals and ground traces, multilayer PCB is a better solution. Alternate layers can be used for signal, ground and power planes—for example, layer 1 as signal plane, layer 2 as ground plane, layer 3 as power plane and layer 4 as another signal plane.

Here is how this concept works: High-frequency current always tries to follow a path that minimises the occupied current loop area because

that path only provides minimum inductance/impedance. All signal traces in the signal plane can use any path below the ground plane for return current. This results in minimum loop area for all signal traces and thus minimum EMI/susceptibility. Signal traces in layer 4 use layer-3 power plane for high-frequency AC return currents. Many grounding and layer stacking schemes are available in literature to take care of these aspects.

Crosstalk

If two signal tracks run very close to each other on a PCB, there will be capacitive coupling between them. High-frequency signal changes on one signal will get transmitted to the other signal because of capacitance. This phenomenon is called crosstalk.

Minimum separation between signal tracks is required to avoid crosstalk. Signal tracks in consecutive layers of a double-sided PCB should be run perpendicularly to avoid crosstalk.

Power supply decoupling

In Fig. 8, assume that the load is the switching load of a digital IC, the source is the power source for that IC and lines shown as transmission line are power supply rails for the IC containing switching load. Normally, the actual power source will be in another power supply card of the unit and voltages will be delivered through connector pins of the card to which this digital IC belongs. Voltages will be delivered to connector pins of the card either through wiring or through a motherboard. Power supply lines from connector pins to the digital IC will run through PCB power tracks. This total current loop will have some inductance because of lengthy power lines and the enclosed large loop area if care is not taken in the PCB layout.

Digital ICs source/receive large switching currents when logic gate circuits containing transistors switch states from high to low or vice versa. If rise/fall time is very low, this switching of current takes place in a very short time. This large change of switching current in a short time results in a voltage drop as per Eq. (3) and voltage spikes appear between

actual power pins of the digital IC. If the total current loop area is large, this can result in EMI also.

Solution to this problem lies in putting power-supply decoupling capacitors between power and ground, one for every digital IC, and at the connector pins where power enters the board. These capacitors will provide the required switching current, without drop in voltage, from their stored charge acquired from power rails. They will regain through power rails the charge lost in supplying the switching current after switching is over. Decoupling capacitors also reduce the effective loop area of switching current since it is supplied by the nearest IC decoupling capacitor. Thus these reduce EMI also.

Usually, 0.1µF ceramic capacitors are used for digital ICs. 10-100µF tantalum capacitors in parallel with 0.1µF ceramic capacitors are used at connector pins where power enters the board. Tantalum capacitors exhibit more inductance at frequencies greater than a certain value, because of which ceramic capacitors are used in parallel to cover a higher frequency range. Simulation tools are available to fix location and values of decoupling capacitors, which need to be used for accurate design.

Knee frequency

If you observe the spectrum of a digital signal on a spectrum analyser, you will find that most of the energy is contained in the frequency band up to a frequency called Knee frequency. The energy rolls off at the rate of -20dB/decade up to knee frequency. Beyond knee frequency, it rolls off much faster. Hence it is considered that most of the energy of a digital signal is contained in the frequency band from 0 to knee frequency, represented as F_{knee} . Value of F_{knee} is related to rise/fall time of digital edges, but not to clock rate.

F_{knee} is given by:

$$F_{knee} = 0.5/T_r$$

where T_r is the pulse rise time/fall time, whichever is lower.

Hence all digital circuits should have a flat frequency response up to F_{knee} to pass digital signal without distortion. **RFY**